GREEN HOME SCORE

AI-Based Green Home Score and Retrofit Recommendation System

TEAM ID: 173

Abstract

The proposed **AI-Based Green Home Score and Retrofit Recommendation System** is an intelligent web platform designed to help homeowners understand the energy efficiency and environmental sustainability of their properties. The system addresses the challenge of identifying how green a home is, what improvements it requires, and how those improvements could reduce energy costs, carbon emissions, and potentially increase property value.

The platform allows users to **enter property details**, including **building age, construction materials, insulation, windows and glazing, heating systems, electrical appliances, renewable energy installations, and existing Energy Performance Certificate (EPC) ratings**, wherever available. Based on these inputs, the system evaluates the property using seven weighted components to generate an explainable **Green Home Score** ranging from 0 to 100. Where supported by available data, the platform can also integrate with smart electricity and gas meters and renewable energy monitoring systems to obtain actual energy consumption and generation data, enabling more accurate and updated assessments.

Using **machine learning models** and documented calculation methods, the system identifies energy inefficiencies and generates personalised, prioritised **recommendations for suitable retrofits**, such as improved insulation, energy-efficient heating systems, double-glazed windows, solar panels, and smart energy controls. For each recommended improvement, the platform **estimates potential energy savings, carbon emission reductions, and financial savings. It also provides an indicative range of potential property value** improvement based on relevant property and market data. An integrated generative AI assistant explains the results in simple language and supports what-if analysis, allowing users to explore how proposed improvements could affect their home's performance before making decisions.

1. Proposed Approach

Property assessment. Collect property characteristics, available EPC information, and—where supported—meter or renewable-generation data. Validate required fields, units, and missing values.

Green Home Score. Calculate an explainable 0–100 score from seven weighted components: energy efficiency (25%), energy consumption intensity (20%), insulation (15%), heating system (15%), windows and glazing (10%), renewable energy (10%), and smart energy management (5%). Each component uses a 0–100 sub-score; the final score is the weighted sum.

Retrofit evaluation. Identify weaknesses and rank feasible measures such as insulation, efficient heating, double glazing, solar panels, and smart controls. Compare baseline and retrofit scenarios with EnergyPlus where suitable models and inputs are available.

Impact and explanation. Use documented calculations and suitable emissions factors to estimate energy, carbon, and financial impacts. Show property-value impact only as an indicative range when relevant market data supports it. An LLM explains structured results and supports what-if questions; it does not originate the underlying numerical results.

2. Proposed System Architecture & Pipeline

PROPERTY INPUTS
Property details + EPC / energy meter data

↓

WEB APPLICATION
Next.js interface → Next.js API → Supabase

↓

SIMULATION
Python assessment worker → baseline model / EnergyPlus retrofit scenarios

↓

SCORING & RANKING
Green Home Score + energy / carbon / cost calculations + retrofit ranking

↓

EXPLANATION
Structured results → LLM explanation API

↓

USER OUTPUT
Interactive dashboard + reports + what-if analysis

Technology stack. Next.js, React, TypeScript and Tailwind CSS; Supabase PostgreSQL; Python and EnergyPlus; server-side LLM API. Suggested hosting: Vercel for the web application and a container host for the simulation worker.

3. Planned Testing Methodology

- **Input validation:** test missing, invalid, out-of-range, and inconsistent property or energy data.
- **Score verification:** check component scores, weights, total score, and category boundaries against hand-calculated cases.
- **Scenario comparison:** verify baseline-versus-retrofit assumptions, units, and energy, emissions, and cost calculations.
- **Recommendation evaluation:** test feasibility filters, ranking logic, and unsuitable or incomplete-input cases.
- **LLM grounding:** compare explanations with structured results and flag unsupported figures or claims.
- **End-to-end and failure tests:** test property submission through results display, including API errors, simulation failures, timeouts, and unavailable integrations.